**5** Fig. 5.1 shows a road junction, a moving car and a stationary truck. The road has high walls on each side.

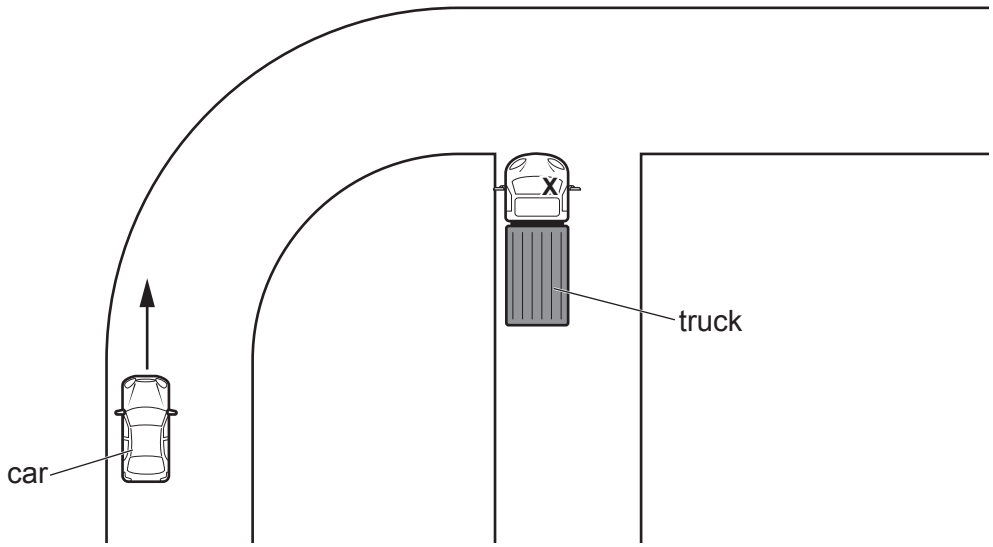



**Fig. 5.1**

**(a)** The driver of the truck is at position **X**. The car moves around the corner.

On Fig. 5.1, label a point Y on the road where the truck driver first sees the car. [1]

**(b)** A plane mirror is placed at the road junction as shown in Fig. 5.2.

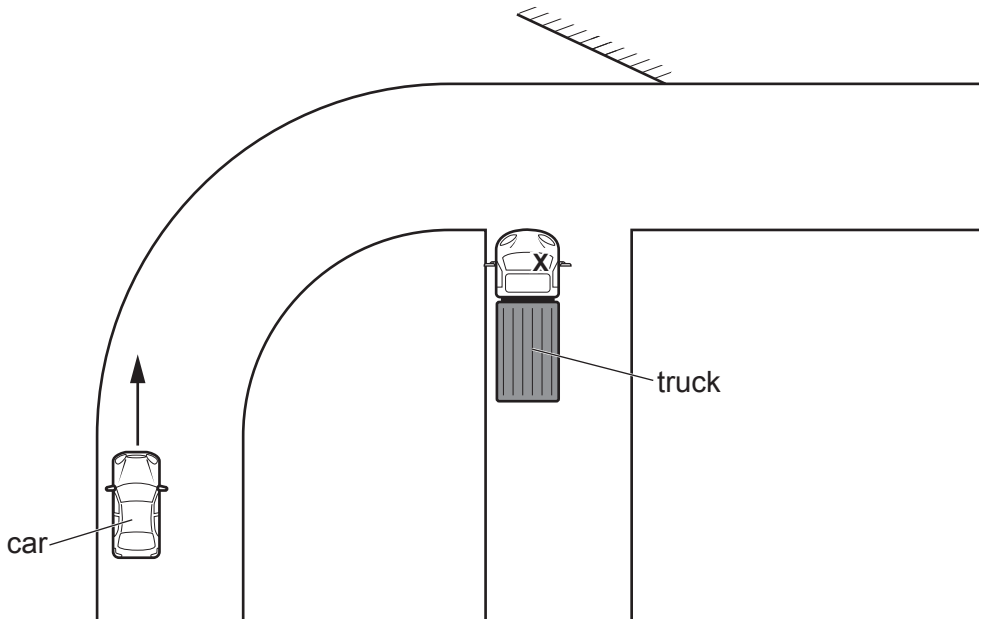



**Fig. 5.2**

Show how this mirror allows the driver of the truck to see the car when it is at the position shown in Fig. 5.2. [2]

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**(c)** The truck driver wears spectacles to correct long-sightedness. Fig. 5.3 shows how a blurred image of an object O forms on the retina. Any effect of the cornea on the rays of light can be ignored.

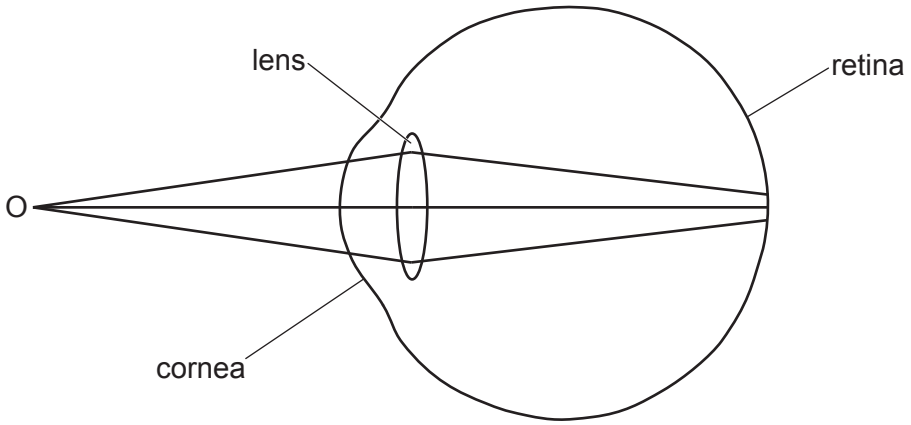



**Fig. 5.3**

On Fig. 5.4, show how long-sightedness is corrected by:

- adding a suitable lens in front of the eye
- continuing the path of the **three** rays of light until they meet to form an image.

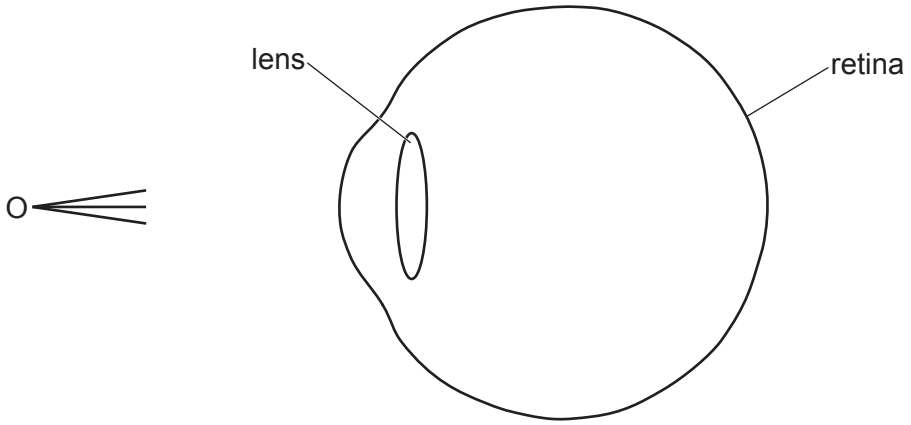



**Fig. 5.4**

[4]

[Total: 7]